

Teaching Dossier

ALEKSA VUJIČIĆ

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1 Teaching Philosophy

The most memorable chocolate I have ever received was after my first job as an instructor, when a student gifted me a bag of KitKat Minis as a thank you for helping them with the course. Of course, it wasn't the gift itself that was important (though it was tasty), but rather that it was an acknowledgement that I had a meaningful impact in a student's education.

This is the goal I still strive for today. When a student engages in mathematical learning, it is rarely the case that they find it easy. I remember this as a student myself - the many hours of being frustrated and confused, trying to understand some mathematical concept. But I also remember the moment I finally found and understood the solution. They were more moments of pure joy, and it was these moments that drove me to keep studying mathematics. Now, whenever I receive the opportunity to teach, I am eager to share this joy with students, so that they may also appreciate the beauty of solving a mathematical problem.

Having had the opportunity to teach a few times, I have found certain aspects to be important to the manner in which I teach. These ideas are certainly not new, and there are of course other facets to my teaching, but those that I give below I believe to be particularly notable to my style of teaching.

To start with, I believe that repetition is important. This does not necessarily mean repeating the same words verbatim, as there are many different forms to repetition. For instance, summarising a previous or current lecture with a few key ideas, or solving multiple problems using a similar method. I believe seeing the same concept multiple times is a very effective method for learning.

I also like to teach using examples. Mathematics has an (arguably deserved) reputation of being incredibly abstract, but even the most intangible theorems are rooted in an observed pattern of examples. As a result, I strongly believe that this is the way we should teach: example first. As teachers, we know the material beforehand, so there is a temptation to do this backwards; to present the theorem and then give examples, as this often feels more elegant.

A classic instance of this is in group theory: the definition of a group a (relatively) simple one, and yet it so elegantly summarises a host of examples observed in many different fields. Of course, as teachers, we want to share this amazing discovery with our students; however, without underlying examples, this beauty is lost. This theory then feels distant and confusing to students. Even when the aforementioned examples are introduced, they can often feel ad-hoc and contrived, rather than a part of a cohesive whole.

Let me provide an example from my own experience. In a multi-variable calculus course, I was introducing the change of variables method in higher dimensions. We computed the integral of $x^2 + y^2$ over an annulus centred at the origin in the 'obvious way' using

Cartesian coordinates. This was a rather tedious computation, so I proceeded to ask if there was a better way, perhaps integrating radially instead. With this motivation, I introduced the notion of change of variables, and students were enthused! We solved the same problem using this new method, and it was clearly how this was a useful tool, not just a random ‘trick’ to remember. Two students even came up to me after class and were eager to learn more.

I also believe that no two students learn the same way. So when I teach, I aim to explain the same concept in different ways. This has the added benefit that students receive a wide variety of tools to tackle a problem, and they can use whichever tool suits them or fits the problem the best. For instance, when I was tutoring students who were working on a GCD problem, many were struggling to develop an intuition for what the GCD operation meant. So together, we went over a couple of different ways to think about it. One way involved drawing a lattice and comparing that to set operations; in another we thought about prime factorisations; and in a third we worked directly with the definition. This enabled the students to interpret the problem in many different ways, and each student could employ their own preferred method (or oftentimes, even multiple methods) to solve their assignment questions.

These are some of the pedagogical ideas about mathematical education that I highly value. I believe that examples are necessary to provide a concrete foundation to any abstract reasoning, that examining these examples from multiple angles is necessary for student engagement, and that repetition helps solidify these ideas. I find teaching to be an incredibly rewarding experience, and I hope I am able to share my love of mathematics to as many people as possible.

2 Teaching Experience

2.1 Instructional

Math 136: Linear Algebra 1

Math 237: Calculus 3

2.1.1 Other Lectures

MATH 900 MATH245(?) lectures for ruxandra

Private Tutoring

2.2 Assistant

Vic

UW

Tutoring Center

2.3 Extensional

- Vic High school - DRP - keatsway?

3 Miscellaneous

3.1 Online

A Sample Material

The **coverage** for this assignment is §2.1–2.5, 3.1–3.2 (inclusive). Your solutions should not use material from any later sections. You are allowed to use material from earlier sections.

Problems

Q1. Prove that $\text{Span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\} = \text{Span} \left\{ \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}$ in $\mathbb{R}^3$.

(Remember that to prove two sets are equal, you must show that they are subsets of each other.)

Q2. Let $(1, 1, 0)$ and $(3, -2, 1)$ be two points on a line $\mathcal{L}$ in $\mathbb{R}^3$.

- (a) Find a vector equation for $\mathcal{L}$.
- (b) Find parametric equations for $\mathcal{L}$.
- (c) Determine whether the point $(-1, 4, -1)$ is on $\mathcal{L}$.
- (d) Determine whether the point $(2, 0, 2)$ is on $\mathcal{L}$.

Q3. Let $\mathcal{L}$ be the line $\mathbb{R}^2$ with the following equation: $\vec{\ell} = \vec{u} + t\vec{v}, t \in \mathbb{R}$, where

$$\vec{u} = \begin{bmatrix} 0 \\ 5 \end{bmatrix} \quad \text{and} \quad \vec{v} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}.$$

- (a) Show that the vector $\vec{x} = [4 \ -3]^T$ lies on $\mathcal{L}$.
- (b) Find a **unit** vector $\vec{n}$ which is orthogonal to $\vec{v}$.
- (c) Compute $\vec{y} = \text{proj}_{\vec{n}}(\vec{x})$ and show that this vector lies on $\mathcal{L}$.
- (d) Prove that $\vec{y}$ is the vector on $\mathcal{L}$ with the smallest norm. That is, prove that $\|\vec{y}\| \leq \|\vec{\ell}\|$ for all $\vec{\ell} \in \mathcal{L}$. (Hint: show that you can write the vectors of $\mathcal{L}$ in the form $\vec{\ell} = \vec{y} + s\vec{v}, s \in \mathbb{R}$ and use Q2a from Assignment 1).

Q4. Let $\mathcal{L}_1$ and $\mathcal{L}_2$ be two lines in $\mathbb{R}^3$ with the respective vector equations:

$$\vec{\ell}_1 = \begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, t \in \mathbb{R} \quad \text{and} \quad \vec{\ell}_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}, t \in \mathbb{R}.$$

- (a) Find a point A which lies on the intersection of $\mathcal{L}_1$ and $\mathcal{L}_2$.
- (b) Let $\mathcal{P}$ be a plane which contains both $\mathcal{L}_1$ and $\mathcal{L}_2$. Find a vector equation for $\mathcal{P}$.
- (c) Find a scalar equation for $\mathcal{P}$.

- Q5.** (a) Determine all real values a and b such that $\begin{bmatrix} 1 \\ 3a \end{bmatrix} \notin \text{Span} \left\{ \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \begin{bmatrix} b \\ 4 \end{bmatrix} \right\}$ in $\mathbb{R}^2$.
- (b) Determine the solution set, S , to the following system of linear equations.

$$\begin{array}{rrrrrcl} 2x_1 & -x_2 & +2x_3 & +4x_4 & = & 0 \\ 2x_1 & -x_2 & & +3x_4 & = & 0 \end{array} .$$

Express S as the span of one or more vectors.

The **coverage** for this assignment is §4.5–4.6, 5.1–5.7 (inclusive). Your solutions should not use material from any later sections. You are allowed to use material from earlier sections.

Problems

Q1. Let $A \in M_3(\mathbb{R})$ be the following matrix:

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & -1 & 0 \\ 1 & 2 & 1 \end{bmatrix}.$$

- (a) Compute the inverse A^{-1} .
 - (b) For a fixed $\vec{b} \in \mathbb{R}^3$, how many solutions does the system $A\vec{x} = \vec{b}$ have?
 - (c) Solve the system $A\vec{x} = \vec{c}$ where $\vec{c} = [1 \ 1 \ 1]^T$. Do not use the Gauss-Jordan algorithm.
- Q2.** Let $A \in M_n(\mathbb{F})$, and suppose that $A^3 + 2A^2 - A + I_n = \mathcal{O}_n$. Prove that A is invertible. What is its inverse?
- Q3.** For each of the following transformations, either prove that it is linear, or find a counterexample to show it is not.

- (a) The transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x+y \\ xy \end{bmatrix}$.
- (b) The transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 3x+2y+z \\ x+z \end{bmatrix}$.
- (c) The transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x-y \\ -x \\ 2+3y \end{bmatrix}$.
- (d) For a fixed $\vec{z} \in \mathbb{R}^3$, the transformation $T_{\vec{z}}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T_{\vec{z}}(\vec{x}) = \vec{z} \times \vec{x}$.

Q4. (a) Determine the kernel of the linear transformation $T_1: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$T_1\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + 2x_2 + 3x_3 \\ x_2 + 2x_3 \\ x_3 \end{bmatrix}.$$

- (b) Is T_1 one-to-one? Justify your answer.
- (c) Determine the range of the linear transformation $T_2: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by

$$T_2\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 \\ x_1 + x_2 + x_3 \end{bmatrix}.$$

- (d) Is T_2 onto? Justify your answer.

Q5. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Suppose that there exists a map $S : \mathbb{R}^m \rightarrow \mathbb{R}^n$ with the property that that

$$S(\vec{x}) \cdot \vec{v} = \vec{x} \cdot T(\vec{v})$$

for all $\vec{x} \in \mathbb{R}^m$ and $\vec{v} \in \mathbb{R}^n$.

- (a) Prove that S is a linear transformation.
- (b) Since both T and S are linear, we can find matrices A and B such that $T = T_A$ and $S = T_B$. Prove that $B = A^T$.

19. A metal rod of length L has its temperature at both ends held at 0°C . The temperature u (in degrees Celsius) at any point x along the rod at time t is given by the equation

$$u(x, t) = \sin(kx) e^{-mt} \quad (1)$$

where k and m are constants.

- [3] (a) Find a non-zero value of k which will satisfy the boundary conditions, that is $u(0, t) = 0$, $u(L, t) = 0$. [Note: there are multiple correct choices for k , you need only provide one.]

- [4] (b) It is known that temperature behaves according to the so-called heat equation

$$u_{xx}(x, t) = \alpha u_t(x, t)$$

where α is a constant. Show that with an appropriate choice of m (in terms of k and α), the formula for the rod temperature in (1) satisfies the heat equation.

B Certificates

August 16, 2023

Aleksa Vujicic
Department of Pure Mathematics

Re: New Instructor Foundations Program

Dear Aleksa,

This letter is to recognize your contributions to undergraduate teaching through your participation in the 3 half-day *New Instructor Foundations Program* for graduate student and postdoctoral instructors offered by the Centre for Teaching Excellence at the University of Waterloo from August 1-3, 2023. This 15-hour program, consisting of 10 hours of in-class time and 5 hours of out-of-class work, is focused on designing and teaching a post-secondary course.

By participating in this program, you have successfully developed your skills in three key areas: course design, assessment design, and peer feedback. Your participation in this program involved:

- aligning your course intended learning outcomes, teaching and learning activities, and assessments,
- identifying strategies that can be used to foster an inclusive learning environment and build classroom community,
- refining or designing the course assessment methods,
- enhancing your course outline,
- actively participating in peer review activities by giving and receiving feedback on course design and assessment plans,
- identifying University of Waterloo resources and people to support you in your work as an instructor, and
- building a community of graduate and postdoctoral instructors with your fellow participants.

Please consider this letter a verification of your completion of the New Instructor Foundations Program. On behalf of the University, I would like to thank you for your commitment to developing your teaching skills, and I congratulate you for completing this program.

Sincerely,



Director, Centre for Teaching Excellence
University of Waterloo



C Course Evaluations

Score Averages

	Question	Mean	SD	Median	Average	Overall Average
Implementation	The instructor(s) helped me to understand the course concepts.	4.1	0.7	4	4.2 SD 0.2	4 SD 0.3
	The instructor(s) created a supportive environment that helped me learn (Supportive environments enable students to feel included and valued regardless of any aspect of their identity).	4.4	0.8	5		
	The instructor(s) stimulated my interest in this course.	4	0.9	4		
Design	The intended learning outcomes were identified (Learning outcomes/objectives articulate what students should be able to know, do, and/or value by the end of a course).	3.9	1	4	3.8 SD 0.3	
	The course activities prepared me for the graded work.	3.4	1.1	3.5		
	The intended learning outcomes were assessed through my graded work.	4.1	0.9	4		
Other	The course workload demands were...	3.7	0.6	4	-	

Figure 1: Caption

Score Averages

	Question	Mean	SD	Median	Average	Overall Average
Implementation	The instructor(s) helped me to understand the course concepts.	4.4	1.2	5	4.3 SD 0.3	4.4 SD 0.2
	The instructor(s) created a supportive environment that helped me learn (Supportive environments enable students to feel included and valued regardless of any aspect of their identity).	4.6	0.8	5		
	The instructor(s) stimulated my interest in this course.	4	1.3	5		
Design	The intended learning outcomes were identified (Learning outcomes/objectives articulate what students should be able to know, do, and/or value by the end of a course).	4.6	0.5	5	4.5 SD 0.1	
	The course activities prepared me for the graded work.	4.4	0.5	4		
	The intended learning outcomes were assessed through my graded work.	4.6	0.5	5		
Other	The course workload demands were...	2.6	0.8	3	-	

Figure 2: Caption

D Teaching Reviews

Peer Teaching Review for Aleksa Vujicic (Fall 2023)

Overview: This review is based on several components:

- 1) An approximately one hour interview via Zoom on Monday, October 2nd.
- 2) A visit to Aleksa's MATH 237 class (Monday, November 13th 2:30-3:20)
- 3) A roughly 30 minute follow-up meeting on Wednesday, November 15th.

The courses being reviewed: This semester Aleksa is teaching MATH 237. MATH 237 is the third course in the Faculty of Mathematics core Calculus sequence. The focus of the course is to provide students with an introduction to the core ideas of Multi-variable Calculus. It covers both the differential and integral aspects of the topic.

Aleksa is teaching one of four sections of MATH 237 this term. The other three are being taught by the course coordinator, a very popular senior lecturer, who is also one of the co-authors of the course notes.

The pre-lecture meeting: The course is very tightly coordinated, with all assessments and all key policy decisions being made exclusively by the course coordinator. On one hand this means less work for Aleksa in delivering the course, but it also means considerably less freedom in terms of how the content is presented and what is to be emphasized. It also means that Aleksa had to do background work to prepare for how the coordinator wanted things done. Based on the interview it seems that he took this quite seriously and before the course began, in fact in July, he went over the course notes to get a feel for what he saw in his own words to be “the big picture” items. The intent was to familiarize himself with key aspects of the course to make links to future lectures when presenting the content. When I asked him what he felt he could add in such a tightly coordinated course, his first reaction was “additional motivation”.

Aleksa did mention that the course uses Piazza as a tool for student questions and that while he was not primarily responsible for monitoring the platform he did so occasionally. Instead he stated that he had regular direct communication with his students through email and via his office hours. Aleksa noted that during his office hours his students would often ask him about things external to the course including more advanced mathematical topics. In particular, he spoke of one student who asked questions about topology. I came away with the impression that Aleksa very much enjoyed these exchanges and encouraged his students to come and see him if they needed help.

Overall, I was quite impressed by what I heard from Aleksa during our meeting. He clearly sees teaching as an important part of his graduate training and possibly his future work after graduation. Aleksa has taken significant steps to prepare himself to teach for the Faculty. In particular, he completed the 1/4-credit graduate teaching training course MATH 900 as well as the Centre for Teaching Excellence's New Instructor Foundations Program in August. In addition to these activities prior to the beginning of the course Aleksa has also taken the time to participate in the Faculty's Teaching Circles Program which sees instructors of various levels of experience visit one another's classes to provide constructive feedback. Moreover, I found him quite animated in speaking about his class. I had a strong sense that he had invested a lot of effort in helping to foster their success.

The lecture: The class started right on time at 2:30 with 19 students out of the 37 registered in attendance. Before commenting further on the lecture it is important to note that this was a terrible room to be assigned to teach this course in particular. It is a large and very badly designed lecture theatre that has the feel of a bomb shelter. There is an upward sliding blackboard that is one modest sized panel

in length. This makes it very difficult to keep reference material in the students sight including the many important diagrams that are such a key part of this course. In front of this very narrow blackboard is a large fixed desk that severely restricts the lecturer's ability to move freely. The acoustics in the room and the lighting are both very bad compared with more modern class rooms. It also did not seem to be set up so that an instructor could use any audio-visual aids in addition to the blackboard. The room was also too large for the enrolment making it challenging to achieve to the type of student engagement that I think Aleksa is capable of.

The topic for the lecture was an introduction to the Change of Variables Theorem for multiple integrals. This is perhaps one of the more difficult aspects of this course and can be quite challenging to teach. This lecture was intended to set up the main theorem which would be presented in a future class. To motivate the topic Aleksa began by reminding the class of an example of a double integral they had seen previously that had been a challenge to evaluate by the basic techniques that they had seen so far. Specifically, they were to evaluate

$$\iint_D x^2 + y^2 dA$$

where D is the annulus $\{(x, y) \mid 1 \leq x^2 + y^2 \leq 2\}$. Aleksa reminded the class that this calculation was long and somewhat challenging and that after much effort they saw that this double integral had value $\frac{15\pi}{2}$. From here he indicated that a strategy to evaluate this integral would be to re-imagine D as all points with polar coordinates (r, θ) where $1 \leq r \leq 2$ and that the general process they want to use is to use "Change of Variables" to transform the integral into one taking advantage of the much simpler description of D using polar coordinates. At this point one of the students asked the question: "Does this have anything to do with substitution?". Aleksa enthusiastically acknowledged this key question with an emphatic YES, and indicated that this was where they would be going in a few minutes. This was one of several instances in the lecture where students provided important contributions. The next example occurred soon after when Aleksa asked his class to supply the resulting integral on the right from the single variable Change of Variables Theorem after the transformation $x = g(u)$

$$\int_a^b f(x) dx \xrightarrow{x=g(u)} \int_{g(a)}^{g(b)} f(g(u))g'(u)du.$$

The next ten minutes of the lecture was a discussion on how to interpret the components of this formula in the setting of double integrals eventually leading the class to the need for a detour to talk about mappings from $\mathbb{R}^n$ to $\mathbb{R}^n$. Throughout the rest of the lecture Aleksa frequently reminded the class of core ideas from earlier in the course concerning differentiability of such functions. He stopped at regular intervals to ask if there were questions and comfortably handled those he received. As was the case earlier in the lecture the students themselves supplied some of the insight Aleksa was trying to highlight including an observation concerning the role to be played by linear approximation and the fact that "if you zoom in enough all nice mappings look linear" as well as confirming a students feeling that a change of basis matrix can be viewed as an important type of mapping.

Throughout the lecture, Aleksa succeeded in keeping the group engaged and actively participating. I did not notice anyone who was not paying attention and there was no noticeable background chatter to distract students in the class. Aleksa's pace was seem relaxed but was very appropriate for the topic. Looking back on my notes, he did cover a lot of material in the lecture. Despite the very poor class room set-up Aleksa's board work was very good. He provided a number of nice illustrative diagrams that would be helpful in understanding the content. His voice was clear and loud enough to be easily heard in the last row of this rather large lecture theatre. Aleksa also made a point of facing the whole class whenever possible while he was speaking or asking for questions.

Post-lecture and the follow-up meeting: The lectured ended on time and immediately several

students approached Aleksa with questions from the lecture. Aleksa spent several minutes answering questions before moving outside the classroom where I had intended to speak to him about the lecture. At that point he indicated that he could not meet me immediately because two students wanted to speak to him about some deeper questions concerning the lecture. I later found out at our follow-up meeting that this was a regular occurrence and that these post-lecture sections could go on for quite sometime. It seems clear that Aleksa is willing to be quite generous with his time.

At the follow-up, Aleksa and I discussed how the lecture went. I mentioned that I had no concerns, but did suggest that when students asked questions or provided answers to his questions it would be a good idea to repeat them to the whole class so that others could hear, and that it would also serve to focus the class' attention on the response. This was particularly relevant in this lecture since the student involvement was on of the strong points. Aleksa was very receptive to this comment and acknowledged that it is something he should concentrate on.

One observation I made was that Aleksa seemed more reserved in the lecture than I had expected, given how enthusiastic he was in our meetings and how engaged he was in answering student questions. I pointed out that I thought that the room might play a role in this. Coincidentally, the day before our follow-up Aleksa substituted for the course coordinator, lecturing in a room with an expansive blackboard and a much better overall environment for this course. He described the experience as "freeing", a word that had come to my mind as well. He felt that being able to simply move around the room made a world of difference.

Summary: Aleksa seems genuinely interested in teaching and obviously put in significant effort to make his students experience a positive one. His technical skills in the classroom are very good and his delivery of the content was clear, easy to follow and appropriate for the course he was teaching. My sense is that Aleksa is very open to constructive comments about his teaching. His participation in Teaching Circles is further evidence of this. For a graduate student teaching a challenging course in an even more challenging environment, Aleksa handled himself very well. I see him as having a strong aptitude for teaching and would be very comfortable recommending that he be given the opportunity to teach additional classes in the future.

Summary: Brian Forrest

December 4, 2023

Observation Report for Aleksa Vujicic

Event Observed: Math 237 (Calculus 3) Lecture
Date Observed: Monday, Oct 30
Location: PAS 2038
Time: 2:30 – 3:20 PM
Number of Students Present: approx 15
Observer: Faisal Al-Faisal

Plan for Teaching Event:

Lagrange multipliers

Aspects to Maintain:

1. Good start to the lecture with a nice example that was grounded in reality (maximize profit).
2. Pace was good and explanations were clear.
3. Voice was audible. Good amount of eye contact and movement around the room. Your tone was friendly and inviting.
4. Very calm and level-headed handling of the disruptions (people walking in multiple times, and once even rudely interrupting you to tell you that this room has been double-booked for a seminar).

Targets for Change and Methods for Improvement:

1. Handwriting was a bit too small. Also be aware of shadows on the board and other potential obstructions.
2. You should always repeat questions/answers from students, as at time what they were saying wasn't audible to me at the back.
3. For a class like this, it might be worthwhile to do an explicit example before starting a proof (e.g. optimize something subject to $x^2 + y^2 = 1$, so that they can conceptualize what is going in—in particular, what is meant by a “parametrization” of the constraint).
4. Try to look for more opportunities for student engagement.

Additional Comments:

Overall, I found the lecture to be well done, despite the less-than-ideal circumstances with the disruptions! You definitely handled that better than I would have.

Note that every successful lesson will demonstrate effective use of many of these elements of excellent teaching, but even the best lessons may not involve them all.

Interactive learning	
☒ Attentiveness to learners' needs	☐ Providing of feedback to learners
☒ Design of activities	☐ Checks for comprehension
☐ Facilitation of activities	☒ Asking of questions to learners
☐ Facilitation of peer-to-peer interactions	☒ Responses to learners' questions
Structure	
☒ Lesson opening or bridge-in	☐ Summary/closure
☒ Communication of learning outcomes	☒ Level of challenge for learners
☐ Pre-assessment	☒ Time management
☐ Post-assessment	
Delivery	
☒ Rapport-building with learners	☐ Use of slides, boards, or other visual aids
☒ Approachability	☐ Use of worksheets or other handouts
☒ Enthusiasm	☒ Vocal pace, volume, articulation
☒ Confidence	☒ Gestures and movement
☒ Clarity of concepts	☒ Facial expressions and eye contact

Observation Report

Event Observed: Aleksa's Math 136 Lecture
Date Observed: March 24
Location: B1 271
Time: 11:30-12:20
Number of Students Present: ~100
Observer: Blake Madill

Plan for Teaching Event: Math 136
8.7 Dimension

Aspects to Maintain:

1. Polling + interacting w/ students
2. board-work, eye contact, + body language
3. enthusiasm

Targets for Change and Methods for Improvement:

1. voice projection (or use mic)
2. holding notes is better than looking at the table.
3. time management + coverage. (~~could~~^{should} go faster)
4. Subscripts too small

Additional Comments:

- Greeted the class + started on time
- Started with motivating example
- Great writing / board work
- Great body language when interacting w/ class + polling
- well prepared
- waited a good "length" for questions.
- could go faster through the proof - too much repetition.
- responds super well to questions + repeats questions

Note that every successful lesson will demonstrate effective use of many of these elements of excellent teaching, but even the best lessons may not involve them all.

Interactive learning	
☒ Attentiveness to learners' needs	☐ Providing of feedback to learners
☒ Design of activities	☒ Checks for comprehension
☐ Facilitation of activities	☒ Asking of questions to learners
☐ Facilitation of peer-to-peer interactions	☒ Responses to learners' questions
Structure	
☒ Lesson opening or bridge-in	☐ Summary/closure <i>Rushed</i>
☒ Communication of learning outcomes	☒ Level of challenge for learners
☒ Pre-assessment	☐ Time management <i>NI</i> ★
☐ Post-assessment	
Delivery	
☒ Rapport-building with learners	☒ Use of slides, boards, or other visual aids
☒ Approachability	☐ Use of worksheets or other handouts
☒ Enthusiasm	☐ Vocal pace, <u>volume</u> , ^{NI} articulation ★
☒ Confidence	☒ Gestures and movement
☒ Clarity of concepts	☒ Facial expressions and eye contact

NI = Needs Improvement.

E References